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Midlife Cardiorespiratory Fitness and Healthy Aging

An Observational Cohort Study

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ABSTRACT

BACKGROUND Higher cardiorespiratory fitness (CRF) is associated with a lower risk for chronic disease and death, but its relation to healthy aging more broadly remains understudied.

OBJECTIVES The aim of this study was to examine the associations between midlife CRF and later life health span (years without major chronic disease), disease burden, and lifespan among adults who remained apparently healthy through 65 years of age.

METHODS This cohort study included 24,576 participants (25% women) from the CCLS (Cooper Center Longitudinal Study) (1971-2017) linked to Medicare administrative claims (1999-2019). CRF was estimated using a maximal treadmill test at a preventive medicine clinic visit before age 65 years. Eleven major chronic conditions were identified from the Medicare Chronic Conditions Data Warehouse and used to define disease as: 1) any of the 11 conditions (composite); 2) any condition within a clinical group (cardiovascular, cardiovascular-kidney-metabolic, cancer); or 3) an individual condition. Multivariable illness-death models estimated the likelihood of transitioning between health, disease, and death by CRF level (low, moderate, or high). Model parameters were used to calculate adjusted Aalen-Johansen probabilities and expected times in each state of health, disease, and death; these were then used to calculate expected health span, number of diseases, disease-years, and lifespan by CRF.

RESULTS When disease was defined as any of 11 major chronic conditions, high-fit men had a 2% (95% CI: 1%-2%) longer health span, 9% (95% CI: 1%-17%) fewer diseases, and a 3% (95% CI: 2%-4%) longer lifespan compared with low-fit men, with similar patterns among women. When disease was clinically grouped, higher fit men and women generally had a later onset of cardiovascular, cardiovascular-kidney-metabolic, and cancer outcomes and developed fewer conditions within each group. On average, the onset of each of the 11 chronic conditions occurred at least 1.5 years later among high-fit men and women compared with low-fit individuals. Results were consistent across clinical subgroups defined by clinic visit year (before or after 1990), age (younger or older than 45 years), smoking status (current smoking, nonsmoking, or missing smoking), and weight status according to body mass index (healthy weight and overweight or obese).

CONCLUSIONS Higher midlife CRF was associated with longer health span, lower multimorbidity, and longer lifespan among men and women. (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

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ABBREVIATIONS AND ACRONYMS

BMI = body mass index

CKM = cardiovascular-kidney-metabolic

COPD = chronic obstructive pulmonary disease

CRF = cardiorespiratory fitness

CVD = cardiovascular disease

ICD-9 = International Classification of Diseases-9th Revision

ICD-10 = International Classification of Diseases-10th Revision

TIA = transient ischemic attack

"Everyone wants to live long, but no one wants to get old."[1] This sentiment will grow increasingly relevant as the U.S. population experiences a demographic shift, whereby those 65 years and older are projected to soon outnumber those younger than 18 years for the first time in history.[2] As the population ages, identifying modifiable factors that can extend health span (years lived in good health) and compress morbidity (years lived with disease) will be critical to improving quality of life, reducing caregiver burden, and managing escalating health care costs.[3-5] Reflecting this priority, the American Heart Association's 2030 Impact Goal aims to increase healthy life expectancy by 2 years.[6]

Health span and disease burden may be related to cardiorespiratory fitness (CRF), an integrative marker of physiological reserve, modifiable primarily through physical activity.[7,8] Unlike self-reported physical activity, CRF is objectively measured and is one of the strongest predictors of health.[8-10] To date, however, most CRF research has evaluated mortality or individual disease endpoints only, limiting understanding of how CRF relates to a more comprehensive view of healthy aging. In particular, whether higher CRF in midlife is associated with a delayed onset of major chronic disease and less disease accumulation over time remains unclear.

Thus, the aim of this observational cohort study was to examine the associations between midlife CRF and key components of healthy aging in later life, including health span, number of diseases, cumulative years lived with disease, and lifespan. This broader perspective extends prior single-outcome and mortality-focused analyses to characterize how CRF relates to the timing and burden of chronic disease in older age. We hypothesized that among individuals who remained apparently free of major chronic disease through 65 years of age, those with higher midlife CRF would have a later onset of disease (longer health span), fewer diseases, fewer disease-years (compressed duration of morbidity), and longer survival.

KEY TERMS

Health span: The number of years lived in good health, operationalized as years free from: 1) any of 11 major chronic conditions; 2) any condition within a predefined clinical group (cardiovascular, cardiovascular-kidney-metabolic, or cancer); or 3) an individual chronic condition.

Disease burden: A general term referring to either the number or the duration of chronic diseases in later life.

Cardiorespiratory fitness (CRF): A clinically measured integrative marker reflecting the functional capacity of the cardiovascular and respiratory systems and primarily modifiable through habitual physical activity.

METHODS

STUDY POPULATION. The study was conducted among CCLS (Cooper Center Longitudinal Study) participants who underwent preventive medicine examinations at the Cooper Clinic (Dallas, Texas) from 1971 to 2017 and subsequently enrolled in Medicare fee-for-service from 1999 to 2019. The CCLS is a prospective cohort study whose aim is to understand the role of CRF and other lifestyle behaviors in health and longevity.[11] The CCLS has received ethics approval from the Texas Tech University Health Sciences Center Institutional Review Board (IRB-FY2025-106), and individuals provided written informed consent to participate in the study.

The sample initially included 44,104 CCLS participants with Medicare fee-for-service enrollment between 1999 and 2019. Exclusions based on the clinic visit included participants who were missing CRF (n = 2,273) or covariates (n = 607); were <20 (n = 1) or ≥65 (n = 2,141) years of age at the clinic visit; had a body mass index (BMI) <18.5 kg/m^2 (who were underweight potentially because of underlying disease) (n = 320); or self-reported histories of myocardial infarction (n = 287), stroke (n = 89), diabetes (n = 581), or cancer (n = 1,308). Exclusions meant to ensure that participants were apparently healthy when entering Medicare surveillance included individuals who qualified for Medicare before 65 years (ie, because of disability or end-stage renal disease) (n = 691); had Medicare claims indicating personal histories of cancer (International Classification of Diseases-9th Revision [ICD-9] V codes, International Classification of Diseases-10th Revision [ICD-10] Z codes) (n = 441) or myocardial infarction (ICD-9 code 412, ICD-10 code I25.2) (n = 40) prior to any recorded disease; or developed disease in Medicare but before entering surveillance (eg, before exclusive fee-for-service enrollment) (n = 1,234). Additionally, consistent with National Institutes of Health recommendations for identifying disease onset in Medicare data,[12] an 18-month blanking period between 65 and 66.5 years was implemented to remove those who may have had pre-existing disease at Medicare enrollment. Accordingly, we excluded participants with disease in Medicare before 66.5 years (n = 6,989) and those who exited surveillance before 66.5 years (whose disease status at 66.5 years was unknown) (n = 2,526). The final analytical sample was 24,576 participants (Figure 1).

ASSESSMENT OF CRF. CRF was estimated at the clinic visit, before 65 years of age, with a maximal treadmill test using a modified Balke protocol, as previously described.[13] Participants continued exercising until volitional exhaustion or clinically indicated test termination. On the basis of the final treadmill speed and grade, maximal METs were estimated,[14] with 1 MET equal to 3.5 mL O_2/kg/min.[15] Using this protocol, duration on the treadmill is highly correlated with the direct measurement of maximal oxygen uptake in men ($r = 0.92$)[16] and women ($r = 0.94$).[17] On the basis of their estimated MET values, participants were categorized into age- and sex-specific groups derived from existing, normative CCLS data. These groups are described as low, moderate, and high CRF according to cutpoints associated with morbidity and mortality.[13,18,19]

ASSESSMENT OF MAJOR CHRONIC DISEASE AND MORTALITY. Medicare claims data from 1999 to 2019 were obtained from the Centers for Medicare and Medicaid Services to identify major chronic disease and death after 65 years of age. Major disease was defined using 11 chronic conditions: heart failure, ischemic heart disease, stroke or transient ischemic attack, diabetes, chronic obstructive pulmonary disease (COPD) and bronchiectasis, chronic kidney disease, Alzheimer's disease and related disorders or senile dementia, colorectal cancer, lung cancer, breast cancer (women), and prostate cancer (men). Disease was identified from the Chronic Conditions Data Warehouse, a CMS resource that uses validated, ICD-9- and ICD-10-based algorithms for chronic disease surveillance and provides the date a beneficiary first met the criteria for a given condition.[20]

These 11 conditions were selected for 2 reasons: 1) to align with our previous study of CRF and later life disease burden,[21] with the addition of sex-specific cancers; and 2) to provide a comprehensive assessment of overall health in later life,[8] as these conditions represent major morbidity across multiple organ systems and account for many of the leading causes of hospitalization and death among older adults.[22-25] Importantly, CRF may be related to each of these conditions, in part, through pathways influenced by habitual physical activity, including cardiometabolic health, immune function, chronic inflammation, and hormonal regulation.[8,10]

Given the complexity of operationalizing the transition from a state of health to a state of disease, we considered 3 separate definitions of disease, where participants were considered to be healthy until they developed: 1) any of the 11 chronic conditions (composite); 2) any condition within a clinical group (cardiovascular, cardiovascular-kidney-metabolic [CKM], or cancer) (see Supplemental Table 1 for conditions included within each group); or 3) an individual chronic condition. Of these definitions, the composite outcome represents the most comprehensive assessment of overall health in later life. Within the composite and clinically grouped outcomes, we accounted for the total burden of disease by allowing the accumulation of multiple diseases during follow-up. Date of death was identified from Medicare data. Cause of death was identified from the National Death Index Plus and used to exclude non-disease-related deaths in sensitivity analysis.

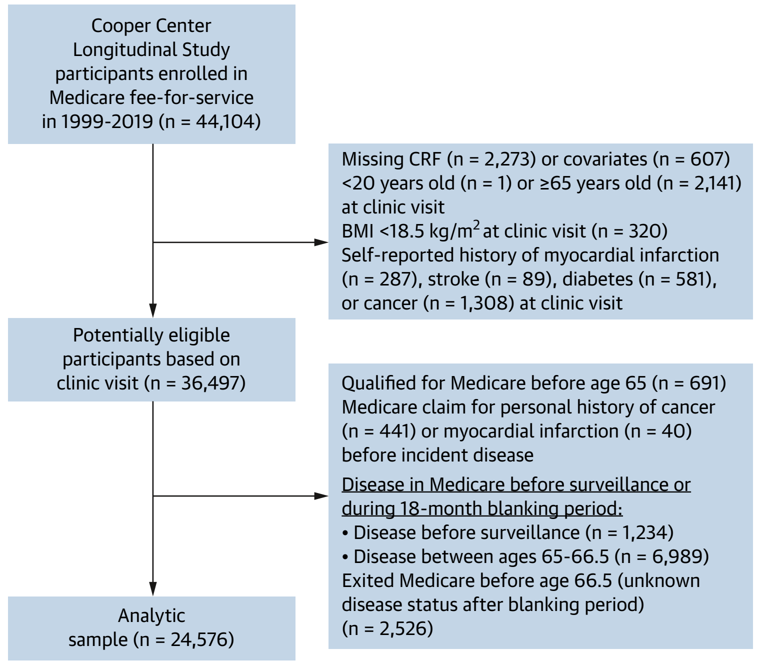
ASSESSMENT OF COVARIATES. Age at clinic visit, year of birth, sex, BMI, current smoking, fasting glucose, total cholesterol, and systolic blood pressure were included as covariates because of their potential associations with CRF[26] and their clinical relevance as risk factors for morbidity and mortality. All covariates were assessed at the clinic visit. BMI was calculated using a participant's weight and height (kilograms per meters squared), which were measured using a standard clinical scale and a stadiometer, respectively.[27] Glucose and cholesterol were measured after a 12-hour fast. Blood pressure was assessed using an established protocol.[27] All covariates were complete except smoking status; missing values for this variable were addressed analytically as detailed in the following section.

STATISTICAL ANALYSIS. Descriptive statistics at midlife clinic visit are presented overall and by CRF. Jonckheere-Terpstra statistics were used to assess trends across CRF levels. Event counts and crude incidence rates of disease by CRF level are provided for the composite, clinically grouped, and disease-specific outcomes. Additionally, unadjusted, nonparametric healthy survival (health span) plots, estimated using the empirical, Breslow method,[28,29] are presented for the composite and clinically grouped outcomes.

Trajectories of aging conditioned on survival to Medicare entry were characterized by 4 semi-competing transition risks: 1) healthy to death; 2) healthy to initial disease; 3) disease to comorbid disease; and 4) disease to death (Supplemental Figure 1). Times to disease and death were expected to be correlated; therefore, an alternative to empirical survival, which assumes that censoring is non-informative, was needed. Additionally, because CRF groups differed on key risk factors (eg, smoking, BMI), adjustment was warranted. Proportional hazards, multitype recurrent-event illness-death models

JACC VOL. ■, NO. ■, 2026
■, 2026:■-■

FIGURE 1 **Study Flow Diagram**



The flow diagram details study exclusions among an initial sample of 44,104 Cooper Center Longitudinal Study participants who were enrolled in Medicare fee-for-service from 1999 to 2019. BMI = body mass index; CRF = cardiorespiratory fitness.

address both of these concerns and were used to estimate mortality parameters, frailty parameter, and hazard ratios (HRs) for CRF (low, moderate, or high) conditional on survival to Medicare entry. Dependence of times to disease and death was modeled by multiplying each transition hazard by a random frailty factor (unit mean) gamma distributed across participants.[30] We assumed Gompertz transition hazards, which are exponential functions of age and well suited to incident mortality and age-related disease.[31,32] Parametric modeling also guaranteed that survival estimates were continuous and smooth (particularly at advanced ages), permitted direct calculation of life expectancy and related quantities, and provided a likelihood function that could be objectively maximized. The maximum likelihood method provides unbiased estimates when data are missing at random without the challenge of defining a compatible imputation model or reliance on random sampling inherent in multiple imputation.[33-35] The handling of tied event times is described in Supplemental Methods 1. Models were sex stratified to account for sex-specific cancers defining major disease and adjusted for year of birth and midlife age, BMI, smoking status, fasting glucose, total cholesterol, and systolic blood pressure. Goodness of fit was confirmed using individual-level, transition-specific Cox-Snell and Martingale residuals.

Model parameters were used to calculate adjusted Aalen-Johansen probabilities[36] for states of health, disease, and death in 0.1-year increments from 65 to 100 years of age by CRF. These were used in turn to calculate expected health span, mean number of diseases, disease-years, and lifespan up to 100 years by CRF, given apparently healthy survival to 65 years. Disease-years are the sum of years affected by each disease, concurrent or otherwise, and account for both the number and duration of each disease. SEs and CIs of health span and related variables were calculated by treating the distribution of model parameters as multivariate normal. Supplemental Methods 2 provides an SAS program that simulates and fits data from the recurrent illness-death model used in the present study. We fit models defining disease as any of the 11 chronic conditions, any condition within a specific clinical group (cardiovascular, CKM, or cancer), or an individual chronic condition.

Heterogeneity in the associations between CRF and aging outcomes was evaluated across a priori

TABLE 1 Characteristics at Clinic Visit (1971-2017) Among Cooper Center Longitudinal Study Participants, Overall and by CRF Level in Midlife

		Men (n = 18,450)			Women (n = 6,126)		
	Overall, 24,576 (100.0)	Low CRF, 2,326 (12.6)	Moderate CRF, 7,347 (39.8)	High CRF, 8,777 (47.6)	Low CRF, 652 (10.6)	Moderate CRF, 2,100 (34.3)	High CRF, 3,374 (55.1)
Race or ethnicity[a]							
White	23,714 (97.6)	2,232 (96.7)	7,083 (97.5)	8,467 (98.1)	621 (96.1)	2,040 (97.4)	3,271 (97.6)
Black	174 (0.7)	29 (1.3)	51 (0.7)	38 (0.4)	11 (1.7)	19 (0.9)	26 (0.8)
Hispanic	175 (0.7)	18 (0.8)	49 (0.7)	50 (0.6)	7 (1.1)	21 (1.0)	30 (0.9)
Other race or ethnicity	232 (1.0)	29 (1.3)	83 (1.1)	75 (0.9)	7 (1.1)	15 (0.7)	23 (0.7)
Current smoking[a]	3,220 (17.3)	663 (37.3)	1,313 (23.9)	808 (12.1)	86 (19.9)	191 (12.8)	159 (5.8)
Age, y	47.6 ± 8.8	45.1 ± 8.4	46.6 ± 8.7	47.9 ± 8.7	46.6 ± 8.7	48.0 ± 9.3	50.3 ± 8.5
Body mass index, kg/m^2	25.7 ± 3.9	29.1 ± 5.0	26.9 ± 3.4	25.2 ± 2.7	26.4 ± 6.1	24.1 ± 4.1	22.6 ± 2.9
Systolic blood pressure, mm Hg	119.8 ± 14.1	123.7 ± 14.2	121.4 ± 13.2	120.9 ± 13.4	117.4 ± 15.6	114.7 ± 14.9	114.5 ± 14.8
Diastolic blood pressure, mm Hg	80.1 ± 9.7	83.6 ± 10.1	81.6 ± 9.5	80.1 ± 9.1	78.3 ± 10.2	77.1 ± 9.8	76.7 ± 9.3
Fasting glucose, mg/dL	98.0 ± 11.8	101.6 ± 17.6	100.0 ± 12.3	98.3 ± 9.9	95.1 ± 10.1	94.2 ± 9.9	93.4 ± 9.2
Total cholesterol, mg/dL	207.7 ± 38.5	215.2 ± 40.2	211.9 ± 39.6	203.9 ± 37.1	208.4 ± 39.4	208.2 ± 39.8	202.7 ± 36
Cardiorespiratory fitness, METs	11.0 ± 2.5	8.5 ± 1.1	10.6 ± 1.1	13.3 ± 1.9	6.5 ± 0.8	8.2 ± 1.0	10.4 ± 1.6

Values are n (%) or mean ± SD. Cardiorespiratory fitness, assessed before 65 years of age, was categorized into age- and sex-adjusted groups from existing, normative Cooper Center Longitudinal Study data. These groups are described as low, moderate, and high CRF on the basis of cutpoints associated with morbidity and mortality. [a]Percentages exclude missing observations. Race or ethnicity was missing for 1.3% of men and 0.6% of women. Current smoking was missing for 24.3% of men and 24.0% of women. Observations with missing smoking status were included in multivariable illness-death models using the maximum likelihood method.

CRF = cardiorespiratory fitness.

subgroups defined by covariates at the clinic visit, including calendar year (<1990 or ≥1990, according to the median clinic visit year), age (<45 or ≥45 years, reflecting the age at which CRF decline typically accelerates[37,38]), smoking status (current smoking, nonsmoking, or missing smoking status), and weight status according to BMI (healthy weight [18.5-<25 kg/m^2] and overweight or obese [≥25 kg/m^2]).

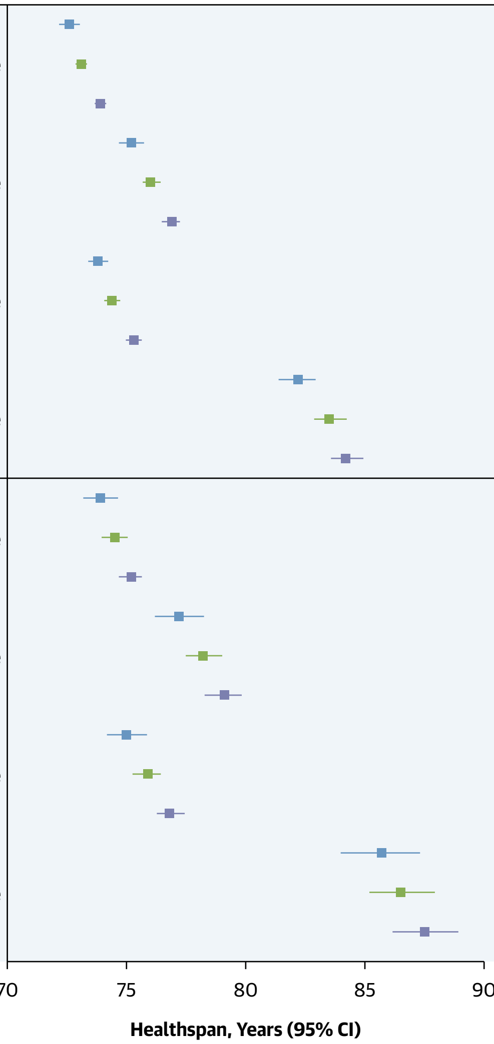
SENSITIVITY ANALYSIS. Four sensitivity analyses were conducted, each within the composite outcome. First, because the study evaluated the relation between CRF and aging among individuals who were apparently healthy when entering Medicare surveillance, we had excluded 10,749 participants with disease in Medicare that occurred either before entering surveillance or before 66.5 years of age (or with unknown disease status at 66.5 years). To evaluate whether findings were robust to the inclusion of these individuals—who were potentially less healthy at entry into Medicare surveillance than the primary analytical sample—we reintroduced these participants into the cohort. Second, we excluded each of the 11 chronic conditions representing disease 1 at a time to assess the sensitivity of results to a particular disease. Third, we excluded the set of cancers (breast, colorectal, lung, and prostate) from the definition of disease. These are cancers for which screening was available during all or part of the follow-up period[39]; screen-detected cancers may be subject to lead-time bias,[40] thus artificially delaying the transition from disease to death and affecting estimates of health span and disease-years. Fourth, we treated non-disease-related deaths as censoring events; this included deaths due to accidents (except falls, which may be disease related), suicide, homicide, legal intervention, and operations of war and their sequelae. All analyses were programmed in SAS/STAT version 9.4 (SAS Institute). Two-tailed *P* values <0.05 were considered to indicate statistical significance, and neither *P* values nor CIs were adjusted for multiple testing.

RESULTS

SAMPLE CHARACTERISTICS. Among 24,576 participants, 75.1% were men and 97.6% were White (Table 1). At the clinic visit, the mean age was 47.6 ± 8.8 years (47.0 ± 8.7 years among men and 49.1 ± 8.9 years among women), and the mean CRF was 11.0 ± 2.5 METs (11.6 ± 2.3 METs among men and 9.2 ± 1.9 METs among women). Participants with higher CRF were more likely to be women and to have a healthier cardiovascular risk profile, including less current smoking and lower BMI, blood pressure, fasting glucose, and total cholesterol (*P* for trend < 0.001 for all). After a mean of 18.9 ± 8.4 years between CRF assessment and Medicare entry and a mean of 10.0 ± 5.9 additional years of

FIGURE 2 Estimated Health Span by CRF Level, Sex, and Disease Outcome Among Cooper Center Longitudinal Study Participants

Sex	Disease Outcome	CRF	Healthspan, Years (95% CI)	Difference, Ratio (95% CI)
Men	Composite	Low	72.6 (72.2-73.0)	Referent
		Moderate	73.1 (72.9-73.3)	1.01 (1.00-1.02)
		High	73.9 (73.7-74.1)	1.02 (1.01-1.02)
	Cardiovascular	Low	75.2 (74.7-75.7)	Referent
		Moderate	76.0 (75.7-76.4)	1.01 (1.00-1.02)
		High	76.9 (76.5-77.2)	1.02 (1.01-1.03)
	CKM	Low	73.8 (73.4-74.2)	Referent
		Moderate	74.4 (74.1-74.7)	1.01 (1.00-1.01)
		High	75.3 (75.0-75.6)	1.02 (1.01-1.03)
	Cancer	Low	82.2 (81.4-82.9)	Referent
		Moderate	83.5 (82.9-84.2)	1.02 (1.00-1.03)
		High	84.2 (83.6-84.9)	1.03 (1.01-1.04)
Women	Composite	Low	73.9 (73.2-74.6)	Referent
		Moderate	74.5 (74.0-75.0)	1.01 (1.00-1.02)
		High	75.2 (74.7-75.6)	1.02 (1.01-1.03)
	Cardiovascular	Low	77.2 (76.2-78.2)	Referent
		Moderate	78.2 (77.5-79.0)	1.01 (1.00-1.03)
		High	79.1 (78.3-79.8)	1.02 (1.01-1.04)
	CKM	Low	75.0 (74.2-75.8)	Referent
		Moderate	75.9 (75.3-76.4)	1.01 (1.00-1.02)
		High	76.8 (76.3-77.4)	1.02 (1.01-1.04)
	Cancer	Low	85.7 (84.0-87.3)	Referent
		Moderate	86.5 (85.2-87.9)	1.01 (0.99-1.04)
		High	87.5 (86.2-88.9)	1.02 (1.00-1.05)



The forest plot shows estimates of health span with 95% CIs by cardiorespiratory fitness (CRF) and sex for the composite outcome and cardiovascular, cardiovascular-kidney-metabolic (CKM), and cancer outcomes. Differences in health span by CRF are presented as ratios comparing participants with moderate or high CRF with those with low CRF. Across each outcome, moderate or high CRF was associated with a 0.5- to 2.0-year delay in the onset of disease, representing a 1% to 3% longer health span. The composite outcome included heart failure, ischemic heart disease, stroke or transient ischemic attack, diabetes, chronic obstructive pulmonary disease or bronchiectasis, chronic kidney disease, Alzheimer's disease and related disorders or senile dementia, colorectal cancer, lung cancer, breast cancer (women), or prostate cancer (men). Clinical groups were defined as cardiovascular (heart failure, ischemic heart disease, or stroke or transient ischemic attack), CKM (heart failure, ischemic heart disease, stroke or transient ischemic attack, diabetes, or chronic kidney disease), and cancer (colorectal, lung, breast [women], or prostate [men]).

follow-up within the Medicare surveillance period, there were 3,358 total deaths (13.7% of the sample), including 2,765 deaths among men and 593 deaths among women (Supplemental Figure 1).

COMPOSITE DISEASE OUTCOME. Crude incidence rates of initial and subsequent major chronic disease decreased with increasing CRF level among both men and women (Supplemental Table 2). Expected health span, disease burden, and lifespan were based on the estimated likelihood of transitioning between health states over time from multivariable illness-death models. Compared with low-fit men, moderate- and

TABLE 2 Estimated Disease Burden and Lifespan by CRF Level and Sex Among Cooper Center Longitudinal Study Participants (Composite Outcome)

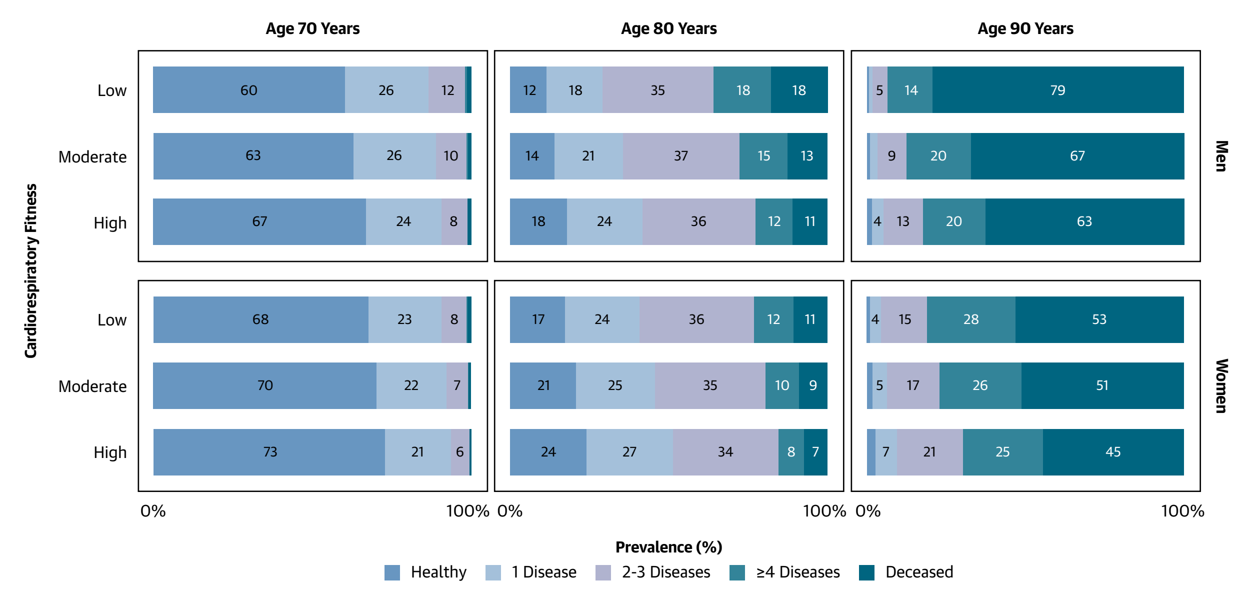
Metrics of Health Given Healthy Survival to 65 Years[a]	Men (n = 18,450)			Women (n = 6,126)		
	Low CRF	Moderate CRF	High CRF	Low CRF	Moderate CRF	High CRF
Mean number of diseases after 65 y	1.7 (1.6-1.8)	1.7 (1.6-1.8)	1.5 (1.4-1.6)	1.7 (1.5-2.0)	1.6 (1.4-1.8)	1.5 (1.4-1.7)
Difference, ratio[b]	Reference	0.99 (0.90-1.09)	0.91 (0.83-0.99)	Reference	0.93 (0.76-1.13)	0.88 (0.73-1.07)
Cumulative disease-years after 65 y[c]	33.7 (30.5-37.3)	36.2 (33.4-39.3)	34.0 (31.5-36.8)	40.9 (33.0-50.8)	38.7 (32.3-46.4)	38.3 (32.2-45.5)
Difference, ratio[b]	Reference	1.07 (0.94-1.22)	1.01 (0.89-1.15)	Reference	0.95 (0.71-1.25)	0.93 (0.71-1.23)
Lifespan, y	85.2 (84.4-85.9)	86.9 (86.2-87.5)	87.5 (86.9-88.1)	88.6 (86.9-90.4)	89.0 (87.5-90.6)	90.0 (88.6-91.5)
Difference, ratio[b]	Reference	1.02 (1.01-1.03)	1.03 (1.02-1.04)	Reference	1.00 (0.98-1.03)	1.02 (0.99-1.04)

Values in parentheses are 95% CI. [a]Estimates of disease burden and lifespan were calculated from illness-death models adjusted for year of birth and midlife age, body mass index, current smoking, fasting glucose, total cholesterol, and systolic blood pressure. Participants were defined as being in a state of health until they developed any of the following major chronic diseases after 65 years of age while enrolled in Medicare: heart failure, ischemic heart disease, stroke or transient ischemic attack, diabetes, chronic obstructive pulmonary disease or bronchiectasis, chronic kidney disease, Alzheimer's disease and related disorders or senile dementia, colorectal cancer, lung cancer, breast cancer (women), or prostate cancer (men). [b]Differences in health metrics by CRF are presented as ratios comparing participants with moderate or high CRF with those with low CRF. For instance, high-fit men were diagnosed with 9% (95% CI: 1%-17%) fewer diseases compared with low-fit men. [c]Cumulative disease-years are the sum of years affected by each disease, concurrent or otherwise, and account for both the number and duration of each disease. For instance, living with two diseases for one year accounts for two disease-years.

CRF = cardiorespiratory fitness.

high-fit men had a 1% to 2% longer health span (Figure 2), corresponding to an additional 0.5 to 1.3 years of healthy life, and a 2% to 3% longer lifespan (Table 2), corresponding to an additional 1.7 to 2.3 years of life. High-fit men also had 9% (95% CI: 1%-17%) fewer diseases after 65 years of age (Table 2). No clear trend in cumulative disease-years by CRF was present. High-fit women had similar point estimates: a 2% (95% CI: 1%-3%) longer health span (Figure 2), corresponding to an additional 1.3 years of healthy life, 12% (95% CI: −7% to 27%) fewer diseases after 65 years of age, and a 2% (95% CI: −1% to 4%) longer lifespan, corresponding to an additional 1.4 years of life (Table 2); however, the only statistically significant difference was the longer health span.

FIGURE 3 Prevalence of a Healthy State, Number of Diseases, and Death by CRF Level, Age, and Sex Among Cooper Center Longitudinal Study Participants



The figure depicts the probability of participants being in a particular state of health, disease, or death at 70, 80, and 90 years of age by cardiorespiratory fitness (CRF) and sex. Probabilities (expressed as percentages and inset by state) were calculated on the basis of multitype recurrent-event illness-death models adjusted for year of birth and midlife age, body mass index, current smoking, fasting glucose, total cholesterol, and systolic blood pressure. At any age, participants with higher vs lower CRF were more likely to be healthy and have fewer diseases and were less likely to be deceased. Overall, 38% of men and 31% of women had Medicare surveillance until death or >80 years, and 21% of men and 14% of women had Medicare surveillance until death or age >90 years. Prevalences not shown are <3%.

TABLE 3 Estimated Disease Burden by CRF Level, Sex, and Disease Outcome Among Cooper Center Longitudinal Study Participants

Metrics of Health Given Healthy Survival to 65 Years[a]	Men			Women		
	Low CRF	Moderate CRF	High CRF	Low CRF	Moderate CRF	High CRF
Cardiovascular (n = 20,880 men, n = 6,850 women)						
Mean number of CV conditions after 65 y	0.7 (0.7-0.8)	0.7 (0.7-0.7)	0.7 (0.7-0.7)	0.7 (0.6-0.8)	0.7 (0.6-0.7)	0.6 (0.6-0.7)
Difference, ratio[b]	Reference	0.98 (0.90-1.07)	0.95 (0.87-1.03)	Reference	0.92 (0.77-1.10)	0.86 (0.72-1.03)
Cumulative CV disease-years after 65 y[c]	14.6 (13.4-15.9)	15.4 (14.3-16.4)	15.4 (14.4-16.5)	16.6 (13.8-20.1)	15.7 (13.5-18.4)	15.3 (13.2-17.8)
Difference, ratio[b]	Reference	1.05 (0.94-1.17)	1.06 (0.95-1.18)	Reference	0.95 (0.74-1.21)	0.92 (0.72-1.17)
CKM (n = 19,675 men, n = 6,575 women)						
Mean number of CKM conditions after 65 y	1.2 (1.1-1.3)	1.2 (1.1-1.3)	1.1 (1.1-1.2)	1.3 (1.1-1.5)	1.1 (1.0-1.3)	1.1 (1.0-1.2)
Difference, ratio[b]	Reference	1.00 (0.92-1.09)	0.93 (0.85-1.01)	Reference	0.90 (0.75-1.08)	0.83 (0.70-1.00)
Cumulative CKM disease-years after 65 y[c]	24.5 (22.3-26.9)	26.3 (24.4-28.4)	25.3 (23.5-27.2)	29.9 (24.5-36.5)	27.5 (23.2-32.5)	26.4 (22.4-31.1)
Difference, ratio[b]	Reference	1.07 (0.95-1.21)	1.03 (0.92-1.16)	Reference	0.92 (0.71-1.19)	0.88 (0.68-1.14)
Cancer[d] (n = 23,745 men, n = 7,127 women)						
Cumulative cancer disease-years after 65 y[c]	2.7 (2.3-3.1)	3.3 (3.0-3.7)	3.7 (3.3-4.1)	2.6 (1.8-3.8)	3.0 (2.3-3.8)	3.1 (2.5-4.0)
Difference, ratio[b]	Reference	1.25 (1.02-1.52)	1.38 (1.14-1.68)	Reference	1.15 (0.74-1.79)	1.20 (0.77-1.86)

Values in parentheses are 95% CI. [a]Estimates of disease burden were calculated from illness-death models adjusted for year of birth and midlife age, body mass index, current smoking, fasting glucose, total cholesterol, and systolic blood pressure. Participants were defined as being in a state of health until they developed any condition within a specific clinical group after 65 years of age while enrolled in Medicare. Clinical groups were defined as cardiovascular (heart failure, ischemic heart disease, or stroke or transient ischemic attack), CKM (heart failure, ischemic heart disease, stroke or transient ischemic attack, diabetes, or chronic kidney disease), and cancer (colorectal, lung, breast [women], or prostate [men]). Sample sizes differ across disease outcomes because participants with diseases other than the disease group of interest prior to entry into Medicare surveillance or within the first 18 months of Medicare enrollment were not excluded from that analysis. [b]Differences in health metrics by CRF are presented as ratios comparing participants with moderate or high CRF with those with low CRF. [c]Cumulative disease-years are the sum of years affected by each disease within a clinical group, concurrent or otherwise, and account for both the number and duration of each disease. For instance, living with two cardiovascular diseases for 1 year accounts for 2 cardiovascular disease-years. There were too few subsequent cancer events after initial events to reliably analyze, so cumulative cancer disease-years represent the number of years living with any one cancer. [d]The mean number of cancers diagnosed after 65 years was not estimated because of a small number of subsequent cancer events.

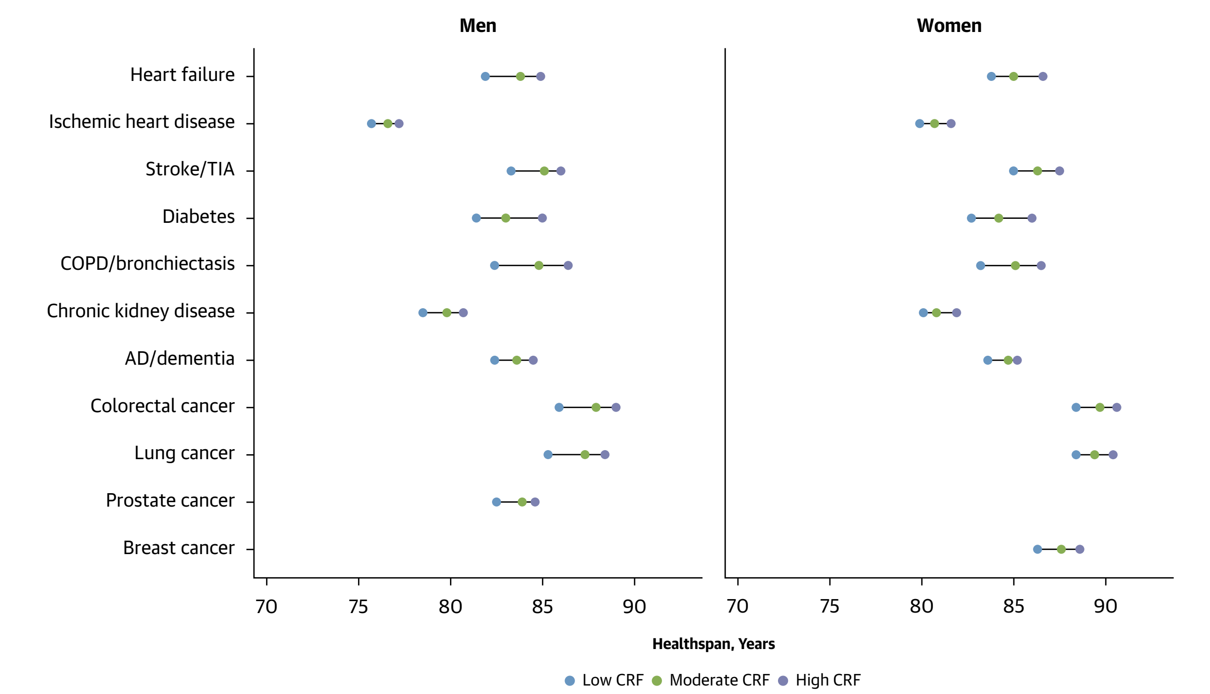
CKM = cardiovascular-kidney-metabolic; CRF = cardiorespiratory fitness; CV = cardiovascular.

Empirical (unadjusted, nonparametric) and modeled (adjusted, parametric) trajectories of health span by CRF level are presented in Supplemental Figure 2, which demonstrate a higher probability of healthy survival (free of any major chronic disease) in later life among those with higher midlife CRF. Additionally, probabilities of being in a state of health, disease, or death at 70, 80, and 90 years of age by midlife CRF level are presented in **Figure 3**. At any age, men and women with higher vs lower midlife CRF were more likely to be healthy and have fewer diseases and were less likely to be deceased.

CLINICALLY GROUPED DISEASE OUTCOMES. Crude incidence rates of initial and subsequent cardiovascular disease (CVD) or CKM disease decreased with increasing CRF level among both men and women. Crude incidence rates of initial cancer also decreased with increasing CRF level (subsequent cancer events were not analyzed because of small sample sizes) (Supplemental Table 2). Associations between CRF and aging metrics within each clinically grouped disease outcome were similar to those observed for the composite disease outcome. Specifically, men and women with moderate or high CRF, compared with low, had a later age at onset of CVD, CKM disease, or cancer (1% to 3% longer health span, corresponding to an additional 0.6 to 2 years of healthy life) (**Figure 2**). High-fit individuals also had fewer CVD and CKM conditions over time, though the differences were generally not statistically significant (**Table 3**). Additionally, moderate- and high-fit men lived 25% to 38% longer after cancer diagnosis, corresponding to an additional 0.5 to 1 year with cancer, compared with low-fit men. Empirical and modeled trajectories of health span by CRF level for each clinically grouped disease outcome are presented in Supplemental Figures 3 to 5, which demonstrate that men and women with higher midlife CRF had a higher probability of healthy survival (free of CVD, CKM disease, or cancer, respectively) in later life.

DISEASE-SPECIFIC OUTCOMES. Crude incidence rates of most individual diseases decreased with increasing CRF level among both men and women; the exception was cancer, which generally had similar incidence rates across CRF level (Supplemental Table 3). Within each disease, men and women with moderate or high CRF had a longer health span compared with those with low CRF (**Figure 4**, Supplemental Table 4). Specifically, high-fit men and women had disease onset that occurred at least 1.5 years later than low-fit individuals; the onset was 3 or more years later for heart failure

FIGURE 4 **Estimated Health Span by CRF Level, Sex, and Individual Disease Among Cooper Center Longitudinal Study Participants**



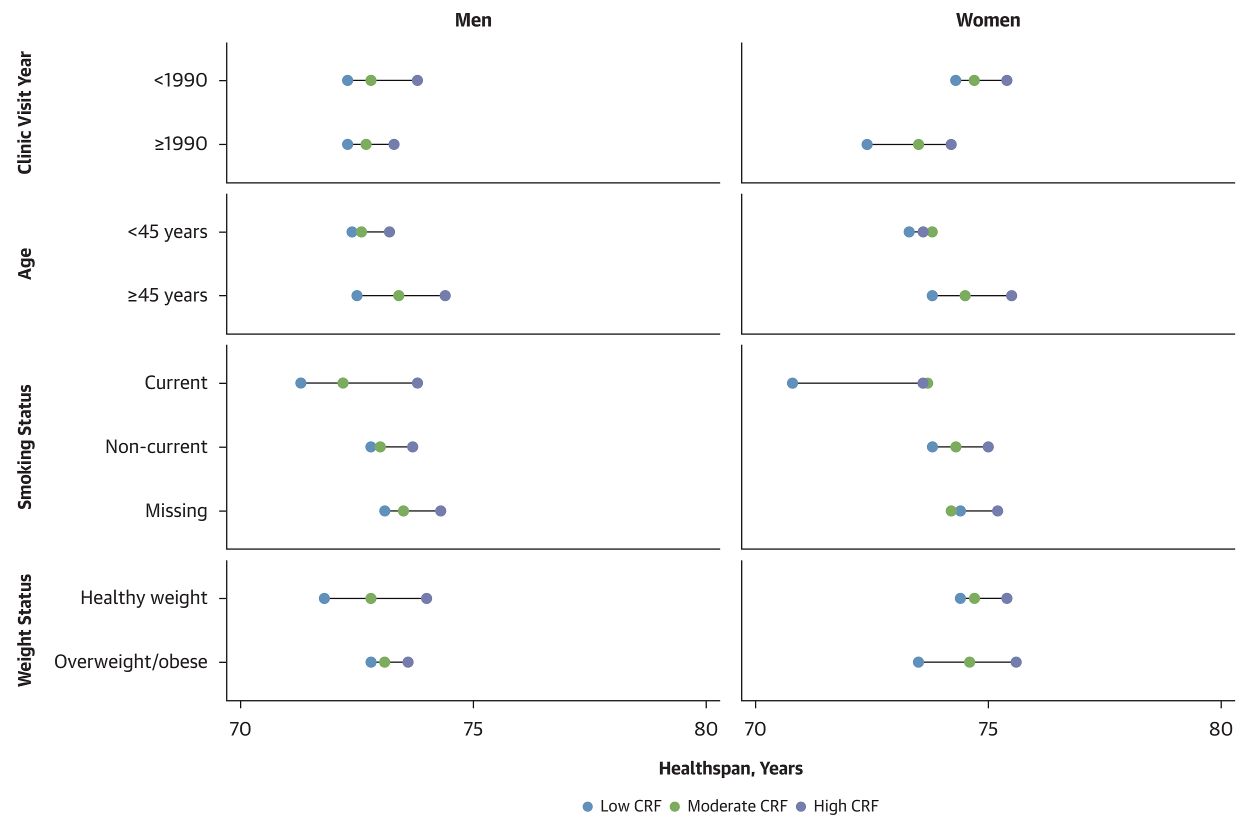
Compared with low cardiorespiratory fitness (CRF), moderate CRF was associated with a 0.7- to 2.4-year delay in the onset of each disease, and high CRF was associated with a 1.5- to 4.0-year delay. AD = Alzheimer's disease; COPD = chronic obstructive pulmonary disorder; TIA = transient ischemic attack.

(men), diabetes (men and women), COPD or bronchiectasis (men and women), colorectal cancer (men), and lung cancer (men). Among men, those with high vs low CRF lived longer with dementia (difference in disease-years, ratio: 1.28; 95% CI: 1.05-1.57) and prostate cancer (difference in disease-years, ratio: 1.32; 95% CI: 1.07-1.64), and they lived fewer years with COPD or bronchiectasis (difference in disease-years, ratio: 0.71; 95% CI: 0.58-0.86). No statistically significant differences were observed by CRF in the number of years lived with each disease among women (Supplemental Table 4).

SUBGROUP ANALYSIS. Within CRF groups (low, moderate, or high), crude rates of initial and subsequent disease were generally higher among men and women with clinic visits prior to 1990, those 45 years or older, those who smoked, and those with overweight or obese BMI (Supplemental Tables 5-8). Nonetheless, in each clinical subgroup, disease rates consistently decreased with increasing CRF level. Similarly, adjusted illness-death models showed no evidence of effect modification: across all subgroups, high-fit men and women were more likely to have a longer health span (**Figure 5**), corresponding to an additional 0.3 to 2.8 years of healthy life, and fewer diseases in later life compared with low-fit individuals (Supplemental Tables 5-8).

SENSITIVITY ANALYSIS. First, to assess whether findings were robust to the inclusion of potentially less healthy individuals, we reintroduced 10,749 participants with disease in Medicare either before entering surveillance or before 66.5 years of age (or with unknown disease status at 66.5 years). Compared with the primary analytical sample, these participants had lower CRF overall and more health-related risk factors at their midlife clinic visits; for instance, the prevalence of current smoking was higher among the excluded group, as were BMI, blood pressure, fasting glucose, and total cholesterol ($P < 0.001$ for all) (Supplemental Table 9). Nevertheless, when the cohort included these participants, higher fit individuals still had a longer health span and lifespan and fewer diseases (Supplemental Table 10). The other 3 sensitivity analyses excluded 1 chronic condition at a time from the definition of composite disease, excluded screen-detected cancers from the definition of composite disease, or treated non-disease-related deaths as censoring events.

FIGURE 5 **Estimated Health Span by CRF Level, Sex, and Clinical Subgroup Among Cooper Center Longitudinal Study Participants**



Compared with low cardiorespiratory fitness (CRF), high CRF was associated with a 0.3- to 2.8-year delay in the onset of major chronic disease across subgroups. Weight status was classified on the basis of body mass index as healthy weight (18.5 to $<$25 kg/m^2) and overweight or obese ($\geq$25 kg/m^2).

Estimates from each analysis were similar to the primary results (data not shown).

DISCUSSION

This cohort study of nearly 25,000 men and women in the United States provides a comprehensive evaluation of the relationship between midlife CRF and key components of healthy aging in later life, including health span, number of diseases, cumulative years of disease, and lifespan. Among individuals who remained apparently healthy through 65 years of age, higher midlife CRF was associated with: 1) later onset of major morbidity, leading to extended health span; 2) development of fewer chronic diseases prior to death; and 3) longer lifespan (**Central Illustration**). These associations were consistent across CVD, CKM disease, cancer, and disease-specific outcomes, as well as across clinical subgroups defined by clinic visit year, age, smoking status, and weight status.

Collectively, these findings have broad implications for clinical practice, public health, and policy and suggest that higher CRF is associated with delayed onset and fewer age-related diseases, including heart disease, cancer, and dementia. Although roughly one-half of the variability in CRF can be attributed to nonmodifiable genetic factors, increases in CRF of 15% to 25%, on average, can be achieved through habitual, moderate- to vigorous-intensity physical activity.[41,42] In our study, more favorable healthy aging metrics were observed for midlife CRF of at least 9 to 12 METs in men and 7 to 9 METs in women, depending on age.

Addressing aging-related disease broadly, rather than preventing or treating single conditions, will be increasingly important for enhancing quality of life in older age and reducing the burden to a health care

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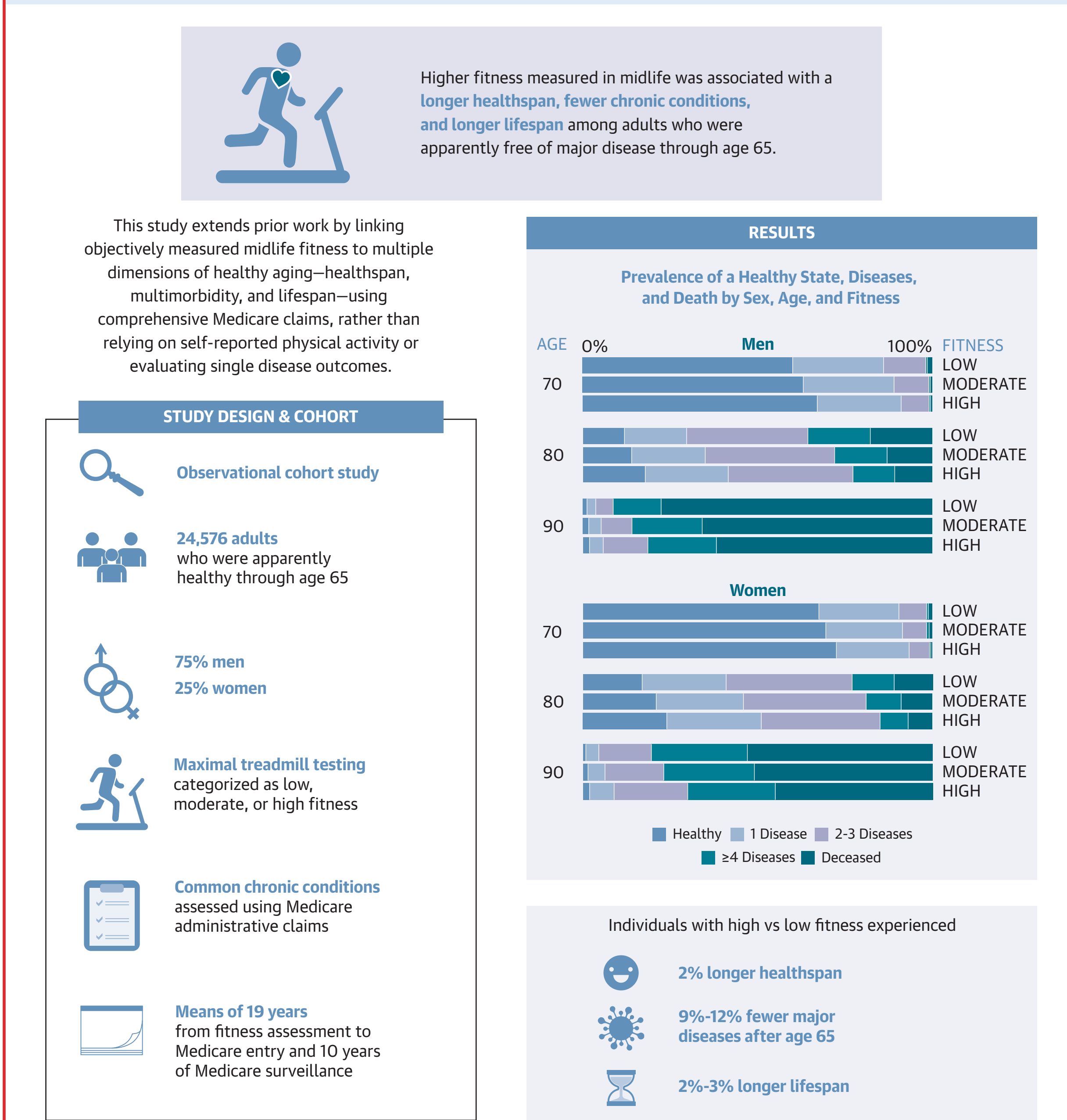
CENTRAL ILLUSTRATION Midlife Cardiorespiratory Fitness and Healthy Aging: An Observational Cohort Study



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system already strained by an aging U.S. population.[5] The economic benefits of delaying aging, that is, simultaneously extending both health span and lifespan by 1 year, as observed among high-fit individuals in this study, are estimated at $38 trillion,[43] roughly equivalent to 8 years of current U.S. health care expenditures.[44] Moreover, in this study, higher CRF was related not only to a later onset of initial disease but also to lower multimorbidity, or the accumulation of multiple chronic diseases. Fewer

JACC VOL. ■, NO. ■, 2026
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diseases may contribute to improved functional status at the individual level,[45] as well as lower health care spending, given that more than two-thirds of health care expenditures in the United States are allocated to patients with multimorbidity.[46]

Although it is well established that higher CRF is related to a reduced risk for morbidity and mortality,[8] previous studies have not concurrently examined health span, disease burden, and lifespan. For instance, an analysis of 5,107 men in the Copenhagen Male Study found that higher levels of CRF, approximately corresponding to moderate and high CRF groups in this study, were associated with lifespans 2.1 and 2.9 years longer than the lowest CRF group, respectively, but other metrics of healthy aging were not examined.[47] Meanwhile, 2 studies among participants in the UK Biobank demonstrated that high CRF was associated with a 10% lower risk for health span termination (defined as the first occurrence of 8 health-deteriorating chronic conditions),[48] a 21% lower risk for multimorbidity, and a 1.3-year later onset of multimorbidity.[49] Similarly, we previously reported that higher fit individuals in the CCLS had a lower risk for chronic disease and were more likely to have fewer diseases at the end of their lives.[21,50] This finding is confirmed in the present study, and with an additional 10 years of Medicare data, we were able to provide estimates of health span, number of diseases, years lived with disease, and lifespan, which were not previously reported.

Longer cancer-free health span and additional years lived after cancer diagnosis among those with higher CRF observed in the present study also aligns with a separate analysis within the CCLS, which showed that higher fit men were less likely to be diagnosed with colorectal or lung cancer and experienced longer survival after colorectal, lung, or prostate cancer compared with low-fit men.[51] In the present study, longer survival after cancer diagnosis among higher fit men appeared to be driven primarily by prostate cancer; future studies should explore whether this reflects a true survival advantage or lead-time bias related to screening.[40]

Collectively, these results suggest that higher CRF in midlife aligns with James Fries's "shift to the right" hypothesis, in which both disease onset and death occur later in life compared with lower fit individuals. Although the present study did not demonstrate a shorter duration of morbidity before death (except for COPD or bronchiectasis among men), it did observe a reduced number of major diseases in later life among those with higher levels of CRF. Indeed, Fries[52] acknowledged that "the national illness burden…may be offset, at least in part, by a lower average illness burden for the individual, with positive consequences for the stability of the health care system." To gain a more comprehensive understanding of morbidity compression, future research should consider the number of diseases accumulated in addition to the time spent living with disease.

STUDY LIMITATIONS. This study was unable to identify disease occurrence during the period between CRF assessment and entry into Medicare surveillance. Nonetheless, linkages between observational cohorts and Medicare claims data with discontinuous surveillance have been frequently used to study the relationship between midlife exposures and later life chronic disease.[21,51,53-55] The length of time between CRF assessment and disease onset minimizes concerns of reverse causality, but it obscures a comprehensive understanding of health at the start of follow-up. Additionally, per recommendations to identify disease onset in Medicare,[12] an 18-month blanking period was implemented at Medicare entry. Although this may have resulted in the exclusion of some participants with true incident disease between 65 and 66.5 years of age, it also limits concerns associated with an "age 65" effect in Medicare, whereby patients delay care for pre-existing disease until they become Medicare eligible.[56] Regardless, results were substantively similar with or without the blanking period exclusion.

Furthermore, results may differ under alternative definitions of disease, such as including other chronic conditions or dimensions of health (eg, physical function or mental health). This analysis focused on conditions that substantially contribute to both hospitalization and death among older adults in the United States.[22-25] At baseline, CRF groups differed significantly on important risk factors (eg, smoking, BMI, glucose). However, our findings remained consistent within clinical subgroups defined by clinic visit year, age, smoking status, and weight status, as well as when modeled to adjust for these differences. Additionally, the frailty factor in the illness-death models accounts for additional sources of individual-level variation that may be unobserved (eg, hereditary).[57] Last, generalizability may be limited given that the cohort was primarily White, had access to preventive health care, was more fit on average than the general U.S. population,[58] and remained apparently healthy through 65 years of age. However, this relative homogeneity may have reduced unmeasured confounding, particularly by social and economic factors that influence physical activity patterns and subsequent CRF.

CONCLUSIONS

These data offer a notable contribution to our understanding of the relationship between midlife CRF and healthy aging in later life. The linkage of objectively measured CRF to long-term Medicare claims data provides a novel and robust data source to ascertain aging-related outcomes. The use of CRF, a physiological consequence of physical activity, is a notable strength compared with previous studies that have used self-reported measures of physical activity.[59-62] Furthermore, prior studies relating physical activity to healthy aging have often been limited to the occurrence of 1 chronic disease, and no consideration has been given to the accumulation of multiple diseases with age.[59-62] The latter limitation is particularly relevant given that 64% of adults in the United States 65 years or older report at least 2 chronic diseases.[63] Finally, this study included a large sample size and robust follow-up, comprising 36,985 disease events and 3,358 deaths. This allowed us to estimate life expectancies with an upper age limit of 100 years, enabling a more accurate representation of health status at the end of life.

These findings provide a foundation for additional research regarding how CRF may influence healthy aging, which can guide clinical practice, public health, and policy. First, more in-depth investigation of the heterogeneity in aging outcomes within CRF groups could help identify subgroups at highest risk for unhealthy aging and support more targeted disease prevention and personalized clinical care. Second, understanding how CRF is related to aging among individuals who develop chronic conditions before 65 years of age is increasingly important given the growing burden of disease at earlier ages.[64] Third, future studies could examine how CRF relates to disease-specific trajectories, including the chronological development of comorbidities (eg, CVD onset after cancer) to clarify biological mechanisms and inform strategies to prevent multimorbidity. Finally, although a single midlife measurement of CRF was related to multiple aging outcomes in this study, it will be important to characterize how changes in CRF over the life course, particularly after initial disease onset, contribute to aging trajectories.

This study revealed that higher CRF is associated not only with a longer lifespan but also with a later onset of major chronic disease and a lower likelihood of accumulating multiple diseases in old age. Among those older than 65 years, it is suggested that the primary goal of medicine and "most important metric of success should be the extension of healthspan."[65] To that end, the identification of a modifiable midlife risk factor such as CRF is promising.

DATA AVAILABILITY The Cooper Center Longitudinal Study (https://www.ttuhsc.edu/centers-institutes/kenneth-h-cooper-institute-.aspx) data are not publicly available. A scientific data request may be submitted to the Kenneth H. Cooper Institute's Scientific Review Board Committee for review. The committee meets regularly to assess the merits of all requests.

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APPENDIX For supplemental tables, figures, methods, and references, please see the online version of this paper.